

Final Database Schema – Expense & Reimbursement System

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1. Roles & User Management

Table: roles

- role_id – Primary Key
 - role_name – Unique, Not Null
 - description
-

Table: admins

- admin_id – Primary Key, UUID
 - full_name – Not Null
 - email – Unique, Not Null
 - password_hash – Not Null
 - phone_number
 - role_id – Foreign Key → roles(role_id)
 - created_at – Default NOW()
-

Table: managers

- manager_id – Primary Key, UUID
- admin_id – Foreign Key → admins(admin_id) (Cascade on delete)
- full_name – Not Null
- email – Unique, Not Null
- password_hash – Not Null
- phone_number

- role_id – Foreign Key → roles(role_id)
 - created_at – Default NOW()
-

Table: departments

- department_id – Primary Key
 - department_name – Not Null
 - description
 - created_at – Default NOW()
-

Table: users

- user_id – Primary Key, UUID
 - manager_id – Foreign Key → managers(manager_id) (Cascade on delete)
 - department_id – Foreign Key → departments(department_id)
 - full_name – Not Null
 - employee_code – Unique
 - email – Unique, Not Null
 - password_hash – Not Null
 - phone_number
 - role_id – Foreign Key → roles(role_id)
 - status – Default 'active'
 - created_at, updated_at – Default NOW()
-

Table: user_activity_logs

- log_id – Primary Key
- actor_id – UUID
- actor_role – Not Null

- activity_type – Not Null
 - description
 - created_at – Default NOW()
-

2. Company Management

Table: companies

- company_id – Primary Key, UUID
 - company_name – Unique, Not Null
 - admin_id – Foreign Key → admins(admin_id)
 - created_at – Default NOW()
-

Table: company_managers

- id – Primary Key
 - company_id – Foreign Key → companies(company_id) (Cascade on delete)
 - manager_id – Foreign Key → managers(manager_id) (Cascade on delete)
 - assigned_at – Default NOW()
 - UNIQUE(company_id, manager_id)
-

Table: company_budgets

- budget_id – Primary Key
 - company_id – Foreign Key → companies(company_id) (Cascade on delete)
 - total_balance – Not Null
 - last_updated – Default NOW()
-

Table: budget_transactions

- transaction_id – Primary Key

- budget_id – Foreign Key → company_budgets(budget_id) (Cascade on delete)
 - reimbursement_id – Nullable, Foreign Key → reimbursements(reimbursement_id)
 - transaction_type – ENUM('Credit','Debit')
 - amount – Not Null
 - performed_by – Foreign Key → admins(admin_id)
 - created_at – Default NOW()
-

3. Expense & Reimbursement

Table: vendors

- vendor_id – Primary Key
 - vendor_name – Not Null
 - vendor_type
 - address
 - created_at – Default NOW()
-

Table: expense_categories

- category_id – Primary Key
 - category_name – Not Null
 - description
-

Table: expense_subcategories

- subcategory_id – Primary Key
 - category_id – Foreign Key → expense_categories(category_id)
 - subcategory_name – Not Null
 - description
-

Table: receipt_types

- receipt_type_id – Primary Key
 - type_name – Not Null
 - description
-

Table: reimbursements

- reimbursement_id – Primary Key, UUID
 - company_id – Foreign Key → companies(company_id) (NEW)
 - receipt_code – Unique, Not Null
 - user_id – Foreign Key → users(user_id)
 - manager_id – Foreign Key → managers(manager_id)
 - admin_id – Foreign Key → admins(admin_id)
 - department_id – Foreign Key → departments(department_id)
 - vendor_id – Foreign Key → vendors(vendor_id)
 - category_id – Foreign Key → expense_categories(category_id)
 - subcategory_id – Foreign Key → expense_subcategories(subcategory_id)
 - receipt_type_id – Foreign Key → receipt_types(receipt_type_id)
 - payment_method – Not Null
 - amount_claimed – Not Null
 - amount_approved
 - currency – Default 'PKR'
 - description
 - ocr_confidence
 - status – Not Null
 - created_at, reviewed_at
-

Table: reimbursement_items

- item_id – Primary Key
 - reimbursement_id – Foreign Key → reimbursements(reimbursement_id) (Cascade on delete)
 - item_name – Not Null
 - quantity – Not Null
 - unit_price – Not Null
 - total_price – Not Null
-

Table: reimbursement_attachments

- attachment_id – Primary Key
 - reimbursement_id – Foreign Key → reimbursements(reimbursement_id) (Cascade on delete)
 - file_name – Not Null
 - file_path – Not Null
 - file_type – Not Null
 - uploaded_at – Default NOW()
-

Table: reimbursement_rules

- rule_id – Primary Key
- category_id – Foreign Key → expense_categories(category_id)
- subcategory_id – Foreign Key → expense_subcategories(subcategory_id)
- max_amount – Not Null
- monthly_limit – Not Null
- requires_manager_approval – Default TRUE
- requires_admin_approval – Default TRUE
- created_by – Foreign Key → admins(admin_id)

- created_at, updated_at – Default NOW()
-

Table: approval_workflow

- approval_id – Primary Key
 - reimbursement_id – Foreign Key → reimbursements(reimbursement_id) (Cascade on delete)
 - approver_id – UUID
 - approver_role – Not Null
 - decision – Default 'Pending'
 - comment
 - decided_at
-

Table: status_history

- status_id – Primary Key
 - reimbursement_id – Foreign Key → reimbursements(reimbursement_id) (Cascade on delete)
 - old_status
 - new_status
 - changed_by – UUID
 - changed_role – Not Null
 - changed_at – Default NOW()
-

Table: policy_violations

- violation_id – Primary Key
- reimbursement_id – Foreign Key → reimbursements(reimbursement_id) (Cascade on delete)
- violation_type – Not Null

- remarks
 - detected_at – Default NOW()
-

4. OCR Insights

Table: ocr_extractions

- ocr_id – Primary Key
 - reimbursement_id – Foreign Key → reimbursements(reimbursement_id) (Cascade on delete)
 - raw_text
 - extracted_json
 - confidence_score
-

5. System & Logging

Table: notifications

- notification_id – Primary Key
 - user_id, manager_id, admin_id – UUID
 - message – Not Null
 - type – Not Null
 - status – Default 'Unread'
 - created_at – Default NOW()
-

Table: audit_logs

- audit_id – Primary Key
- actor_id – UUID
- actor_role – Not Null
- action – Not Null

- entity – Not Null
- entity_id – UUID
- old_data, new_data – JSONB
- created_at – Default NOW()